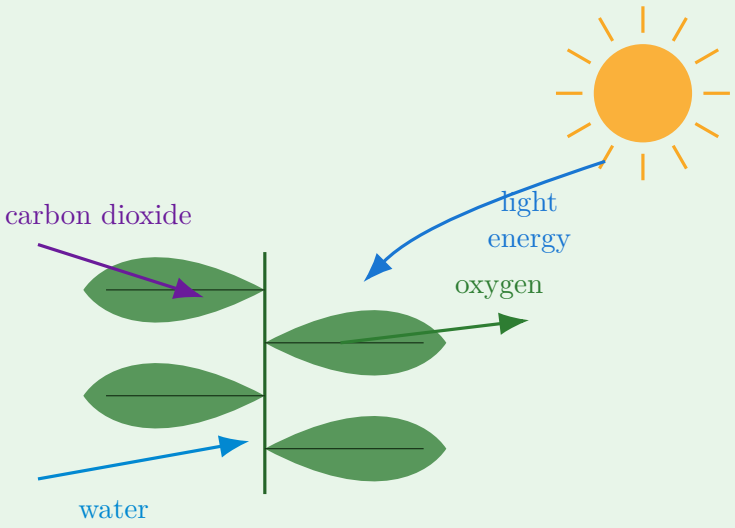
PHOTOSYNTHESIS

Student-Friendly Notes

Class 7 Science — Life Processes in Plants



$$\mathbf{CO_2 + Water \xrightarrow[chlorophyll]{sunlight} Glucose + O_2}$$

Built from the uploaded Grade 7 *Curiosity* textbook, Chapter 10.
Main teaching focus: **photosynthesis**. For completeness, **all 10 end-of-chapter exercise questions are answered**.

Includes diagrams, experiments, memory tricks, common mistakes, exam answers, and a quick revision sheet.

Editable Overleaf project — all diagrams are drawn in TikZ, so no external image files are needed.

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All Textbook Exercise Questions — Solved

The teaching focus stays on photosynthesis, but all 10 questions under Let Us Enhance Our Learning *are answered so the handout is complete. Questions 5 and 9 are clearly marked as chapter-context questions.*

Q1 — Complete the table: Photosynthesis column

Raw materials: carbon dioxide and water.
Products: glucose and oxygen.
Word equation: Carbon dioxide + Water $\xrightarrow[\text{chlorophyll}]{\text{sunlight}}$ Glucose + Oxygen.
Importance: prepares food for the plant and releases oxygen; it is a major source of food and oxygen for living organisms.
Tip: Separate *raw materials* (CO_2, water) from *conditions* (sunlight, chlorophyll).

Q2 — What if all photosynthetic organisms disappeared?

Exam-ready Answer

Food production at the base of most food chains would stop. Herbivores would lose their food source, followed by carnivores and other consumers. Oxygen released by photosynthesis would no longer be replenished in the same way, while carbon dioxide would continue to be produced by respiration and other processes. Therefore, the balance of gases and the survival of most living organisms would be severely affected.

Memory tip: Think **FOOD + OXYGEN + BALANCE**.

Q3 — Where does the starch in a potato come from?

Exam-ready Answer

Food is synthesised mainly in the green leaves by photosynthesis. The leaves make glucose, some of which is converted to stored food. Food is transported from the leaves to other parts of the plant through **phloem**. In potato, it reaches the underground stem (tuber) and is stored there largely as starch. Therefore, the starch detected in a potato ultimately comes from food made by photosynthesis in the leaves.

Common mistake: A potato does not obtain its starch directly from the soil.

Q4 — Why are broad, flat leaves efficient for photosynthesis?

Exam-ready Answer

A broad and flat leaf has a large surface area exposed to light. This helps it capture more sunlight. Leaves also contain chlorophyll and have stomata for gas exchange. These features help the leaf carry out photosynthesis efficiently.

Tip: For a short answer, write: **large surface area → more light captured**.

Q5 — Identify X, Y and Z in the respiration equation

The equation is:

$$X + Y \longrightarrow \text{Carbon dioxide} + Z + \text{Energy}$$

From the chapter's respiration equation, **X = Glucose, Y = Oxygen, and Z = Water.**
Why this is included: it is a respiration question from the same Chapter 10 exercise, even though the main notes are about photosynthesis.

Q6 — One plant in sunlight, one in complete darkness

(i) Idea being tested: Whether sunlight is necessary for food/starch formation and healthy plant growth.
(ii) Visible difference: The plant receiving sunlight is expected to remain greener and healthier, while the plant kept in complete darkness becomes pale/yellowish and shows poorer growth.
(iii) Iodine test: Leaves from the plant kept in sunlight are expected to show the blue-black starch result; the dark-kept plant should not show this starch result after appropriate destarching/controlled conditions.
Tip: Link the answer chain: **sunlight** → **photosynthesis** → **starch** → **blue-black with iodine**.

Q7 — Four conditions: light/dark and CO_2/no CO_2

For photosynthesis, both **sunlight** and **carbon dioxide** are required (with water and chlorophyll available).

Set-up	Condition	Starch?	Oxygen from photosynthesis?
(a)	Sunlight + carbon dioxide	✓ Yes	✓ Yes
(b)	Sunlight, no carbon dioxide	✗ No	✗ No
(c)	Dark + carbon dioxide	✗ No	✗ No
(d)	Dark, no carbon dioxide	✗ No	✗ No

So: **(i)** starch forms in (a) only; **(ii)** not in (b), (c), (d); **(iii)** oxygen is generated by photosynthesis in (a) only; **(iv)** not by photosynthesis in (b), (c), (d).
Tip: Use an **AND rule**: photosynthesis needs light **AND** CO_2 (plus water and chlorophyll).

Q8 — Snail, water plant, both, and water-only test tubes

The setup is designed to study the relationship between **carbon dioxide from animal respiration** and its **use by a water plant during photosynthesis**. A carbon dioxide indicator changes according to the amount of CO_2 present.

- **A: snail only** — the snail respires and releases CO_2, so the indicator should show a change corresponding to increased CO_2.
- **B: plant only** — in light, the plant uses CO_2 during photosynthesis, so the indicator should show a change corresponding to reduced CO_2.
- **C: snail + plant** — CO_2 released by the snail can be used by the plant; the change may be smaller or tend toward balance, depending on rates.
- **D: water only** — acts as a control for comparison.

She supports her idea by comparing the indicator colour in A, B and C with the control D after the same time and under the same conditions.
Common mistake: The textbook does not specify the particular CO_2 indicator or its exact colour scale here, so do not invent a colour. State the expected *direction of CO_2 change*.

Q9 — Design an experiment: Is water transport faster in warm or cold conditions?

Use **two similar freshly cut twigs** of the same plant and place each in an equal amount of water containing the **same number of drops of red ink**. Keep one set-up in a **warm place** and the other in a **cool place**. Keep other factors as similar as possible.
Observe at fixed time intervals and record either:

- the **time taken** for red colour to first appear at the same height in the stem/leaves/white flower, or
- the **height reached** by the coloured water after the same amount of time.

The condition in which the red colour moves farther in the same time (or reaches the chosen point sooner) has the **faster water transport**. Water moves upward through **xylem**.
Expected trend: transport is generally faster under warmer conditions because water loss from leaves (transpiration) tends to be greater; the experiment should be used to test this expectation.

Q10 — How do photosynthesis and respiration help maintain balance in nature?

Exam-ready Answer

Photosynthesis uses carbon dioxide and releases oxygen while making food. Respiration uses oxygen to break down glucose and releases carbon dioxide, water, and energy. Thus, the products of one process are used in the other. Together, these processes help cycle carbon dioxide and oxygen and support the flow of food and energy through living systems.

Tip: Draw two arrows: **Photosynthesis: $CO_2 \rightarrow O_2$**; **Respiration: $O_2 \rightarrow CO_2$**.

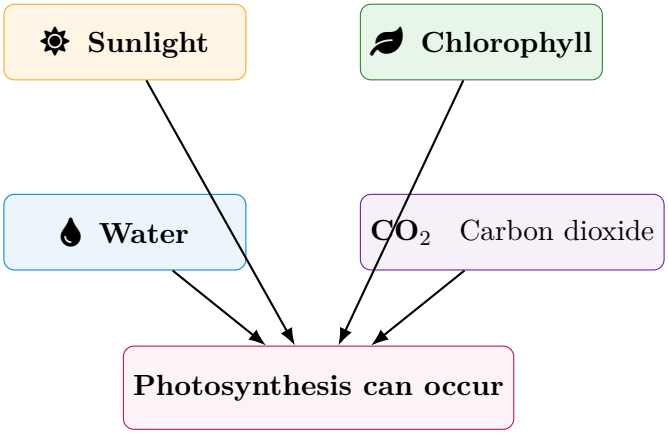
The Big Question: How Do Plants Get Food?

Animals eat food, but plants do not eat in the same way. Green plants **make their own food**. The main food-making process is **photosynthesis**.

Key Idea

Photosynthesis is the process by which green plants use **carbon dioxide** and **water**, in the presence of **sunlight** and **chlorophyll**, to make **glucose**. **Oxygen** is released during the process.

The four essentials



Memory Trick

Think **S-W-C-C**: **S**unlight + **W**ater + **C**arbon dioxide + **C**hlorophyll. If one is missing, normal photosynthesis cannot make starch in the tested leaf.

Where does it happen?

The **leaf** is the primary site of photosynthesis. Leaves are usually broad and flat, helping them receive light efficiently. Their green colour is mainly due to **chlorophyll**, a pigment that helps capture sunlight. Other green parts containing chlorophyll can also photosynthesise.

Common Mistake

Do not write that **roots make food from soil**. Roots absorb water and minerals. The food is synthesised mainly in green leaves.

The Photosynthesis Equation

$$\text{Carbon dioxide} + \text{Water} \xrightarrow[\text{chlorophyll}]{\text{sunlight}} \text{Glucose} + \text{Oxygen}$$

Inputs / raw materials

- Carbon dioxide from air
- Water absorbed by roots

Conditions and products

- Sunlight provides light energy
- Chlorophyll captures light
- Glucose is produced
- Oxygen is released

Glucose is a simple carbohydrate. It can be used as an immediate source of energy and can later be converted into **starch** for storage.

Memory Trick

G comes before S: the plant first makes **G**lucose; some glucose is later stored as **S**tarch. The iodine test detects **starch**, not "photosynthesis" directly.

How Do We Know? — The Key Experiments

These experiments are especially important because many textbook questions ask you to use evidence, not just recall a definition.

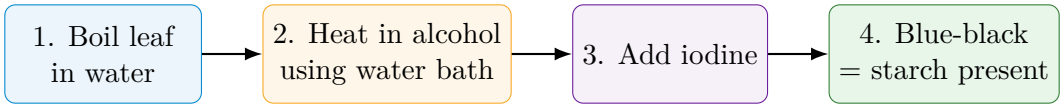
Experiment 1: Iodine test for starch

Activity / Evidence

Purpose: To test whether a leaf contains starch.
Method:

1. Soften the leaf in boiling water.
2. Put the leaf in alcohol kept in a hot-water bath until the leaf loses its green colour.
3. Place the decolourised leaf on a plate and add diluted iodine solution.
4. A **blue-black** colour shows that **starch is present**.

Why remove the green colour? So the iodine colour change can be seen clearly.



Common Mistake

Alcohol is highly flammable. It must **not** be heated directly over a flame. In the textbook activity, it is heated indirectly in a hot-water bath and treated as a teacher demonstration.

Experiment 2: Sunlight and chlorophyll are needed

A leaf with green and non-green patches is useful because different parts of the *same leaf* can be compared.

Plant kept in sunlight **Plant kept in dark**



After iodine: green patches turn blue-black

After iodine: no blue-black starch result

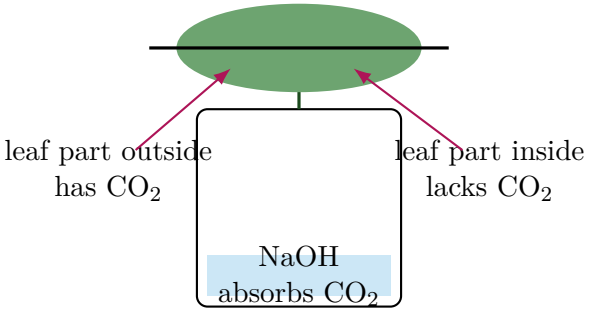
Inference: In the plant exposed to sunlight, starch is detected in the green patches. The leaf from the plant kept in darkness does not show the blue-black starch result. This supports the need for **sunlight** and **chlorophyll** in starch formation.

Key Idea

A non-green patch may have too little chlorophyll to make enough starch for detection. Also remember: some red, violet, or brown leaves may still contain chlorophyll hidden by other pigments.

Experiment 3: Carbon dioxide is needed

A destarched potted plant is used. Half of one leaf is placed inside a bottle containing sodium hydroxide (caustic soda), while the other half remains outside. Sodium hydroxide absorbs carbon dioxide. The setup is kept in sunlight and the leaf is later tested with iodine.



Result: outside part forms starch; inside part does not.

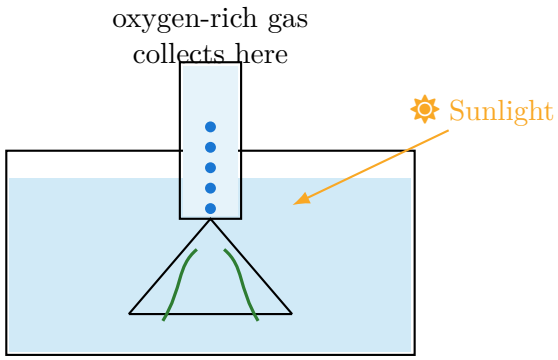
Conclusion: Carbon dioxide from air is essential for plants to prepare starch.

Common Mistake

Do not say sodium hydroxide "gives carbon dioxide". It does the opposite: it **absorbs carbon dioxide** from the air inside the bottle.

Experiment 4: Oxygen is released

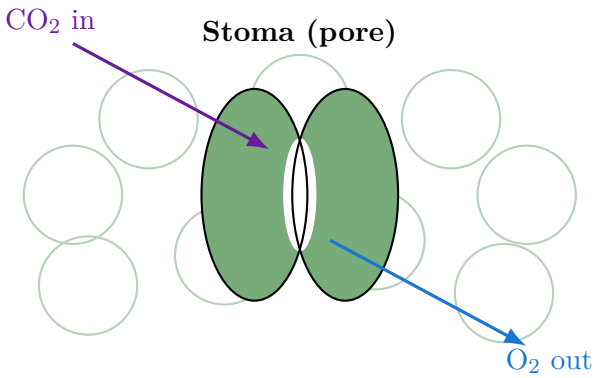
A water plant is placed under a funnel with an inverted test tube. In sunlight, bubbles collect in the tube. The collected gas is rich in oxygen. This supports the conclusion that oxygen is released during photosynthesis.



Water plant under funnel

Gas Exchange: Stomata

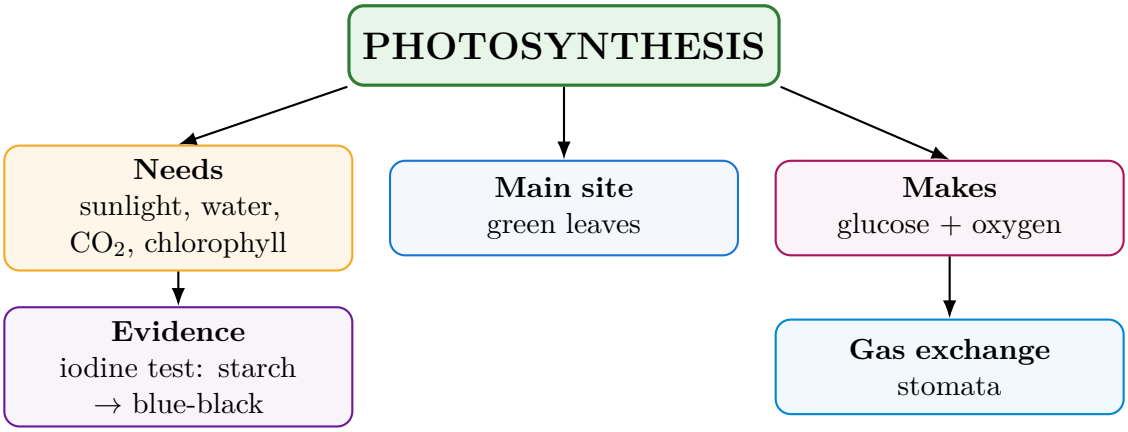
Leaves have tiny pores called **stomata**. They help in the exchange of gases. During photosynthesis, carbon dioxide is needed and oxygen is released.



Memory Trick

STOMATA = tiny "doors" for gases. For Class 7, remember: carbon dioxide enters for photosynthesis and oxygen produced can leave through stomata.

One-Page Concept Map



Exam Toolkit

Very short answers

Photosynthesis	Food-making process in green plants using CO_2 and water in the presence of sunlight and chlorophyll, releasing oxygen.
Chlorophyll	Green pigment that helps capture sunlight for photosynthesis.
Stomata	Tiny pores on leaf surfaces that help in gas exchange.
Iodine test	Blue-black colour indicates starch.
Main site	Leaf; other green chlorophyll-containing parts can also photosynthesise.
Stored food	Glucose can later be converted into starch for storage.

Five common mistakes to avoid

1. ✗ "Plants get food from soil." ✓ They get water/minerals from soil and make food by photosynthesis.
2. ✗ "Sunlight is a raw material." ✓ It is an energy source/condition; the raw materials are CO_2 and water.
3. ✗ "Iodine tests for glucose." ✓ In this chapter, iodine is used to test for starch.

4. ✗ "Oxygen is needed to make glucose in photosynthesis." ✓ Oxygen is released during photosynthesis.
5. ✗ "Only leaves can photosynthesise." ✓ Leaves are the primary site; other green parts containing chlorophyll can also photosynthesise.

How to answer an experiment question

Use the four-word pattern **Aim** → **Setup** → **Observation** → **Conclusion**.

Memory Trick

For any "Is X necessary for photosynthesis?" question, compare a setup where **X is available** with one where **X is missing**, while keeping other important conditions the same. Then use the iodine starch test as evidence when appropriate.

Rapid Revision: 60 Seconds

- **Needs:** CO_2 + water + sunlight + chlorophyll.
- **Makes:** glucose + oxygen.
- **Stores:** some glucose as starch.
- **Main site:** green leaves.
- **Gas doors:** stomata.
- **Starch test:** iodine → blue-black.
- **No light?** No detectable starch formation in the textbook comparison.
- **No CO_2?** No starch formation in the tested leaf part.
- **No chlorophyll?** Insufficient/no detectable starch in the non-green patch under the activity conditions.
- **Why it matters:** makes food and releases oxygen, supporting living organisms and food chains.

Self-Check Quiz

Try these without looking back.

1. Name the two raw materials of photosynthesis.
2. What is the role of chlorophyll?
3. Why is a leaf decolourised before adding iodine?
4. What colour shows the presence of starch?
5. What does sodium hydroxide do in the carbon dioxide experiment?
6. Why does the water-plant setup produce bubbles in sunlight?
7. Why is a broad, flat leaf useful?
8. Through which structures do leaves exchange gases?

Answers — check after attempting

1. Carbon dioxide and water.
2. Chlorophyll captures sunlight.
3. Makes the iodine colour change visible.
4. Blue-black indicates starch.
5. Sodium hydroxide absorbs CO_2.
6. Oxygen is released.
7. A broad leaf collects more light.
8. Gas exchange is through stomata.